

Element B

Once we assessed the problem, we then moved into identifying previous attempts at resolving this issue. In terms of finding solutions for Komodo dragon containment, there was little information available about specific caging methods. However, similar reptiles such as crocodiles and alligators present comparable challenges. Through lots of research, we found several solutions for other animals, but since Komodos are a newer addition to most zoos, there are currently no well-established solutions for this specific problem. This is the common issue we found when looking for possible solutions for our box. Another issue that we ran into was the various requests of the zoo. With these requests, finding previous solutions that accomplished the same requirements was difficult.



The most common crocodile/alligator cage design uses a long, segmented cage that is also collapsible. The purpose of our cage design is to secure the animal while allowing zookeepers to safely perform medical procedures, such as injections and x-rays, without coming into direct contact with the animal. Most often, the cages are extremely long to ensure that any size animal fits inside without restriction.

In researching solutions for similar species, we found that segmented, collapsible cage designs are commonly used for crocodiles and alligators. These systems often feature metal or aluminum frames, ventilated access points, and sliding or guillotine-style doors. One design, used in transport crates at large zoos such as the San Diego Zoo (as documented in veterinary logistics case studies), allows handlers to administer injections and conduct imaging through access hatches without having to come into direct contact with the dragons. The use of reinforced materials, combined with clear windows and built-in ports, improves both safety and medical accessibility

One previously identified solution incorporates both collapsibility and segmentation in its design. The windows in this design are made of a combination of metal, plastic, and elastic materials, allowing for adequate ventilation while also providing access points for handling the animal. The segmentation of the boxes allows for ease of transport and adjustability, making it possible to accommodate animals of varying lengths. This is one of the requirements that we are keeping in mind when designing our box because we have to be able to accommodate for Komodo dragons that are 3, 7, and 9 ft long.



Again, a very similar design to that listed above, however, this design seems much more secure in terms of the materials that were utilized. This design also achieves the goal of ease of transportation and collapsibility while also being fully secured in a metal/aluminum frame. Also, the gate that traps the animal in the cage is vital to the functionality of the overall design. This door is a necessity for our design, however, it will be more complicated because we have segmented boxes so we will have to figure out a way to design it with that in mind.



The current solution used at the Nashville Zoo consists of a large plastic tube. The zookeepers current process requires multiple keepers to transport. However, zookeepers have identified multiple critical flaws in this design. The lack of visibility makes it difficult to monitor the animal, and the absence of access ports significantly complicates medical procedures. Zookeepers report that drawing blood alone can take upwards of 20 minutes, and there is no method for trimming the dragons' nails. Additionally, there are no designated areas for positioning x-ray plates, making imaging procedures nearly impossible. Overall, this design is a makeshift solution that lacks both safety and functionality.

Zookeepers have expressed a clear need for a redesigned containment system that addresses these shortcomings. Due to a lack of time and resources, they have been unable to develop an improved alternative. Our solution must incorporate adjustability, visibility, safety, and accessibility while ensuring ease of transport and easier administration of medical procedures.

Our project aims to build on the strengths of these prior designs such as collapsibility, secure gating, and medical access, while addressing their limitations. We are incorporating a segmented design composed of three 3-foot long modular units to provide flexibility for dragons of varying sizes. We also plan to implement a custom connection system that ensures structural stability and even weight distribution. Because no current system meets these exact requirements, our design introduces a novel approach inspired by prior reptile handling strategies and adapted specifically to the Komodo's unique needs.

The next thing that we had to focus on was creating a connection system so that the komodo's weight would be better distributed and so that the boxes would be stable enough to hold the komodos securely without the box failing to stay together securely. In terms of the research that went into this, there was nothing relevant that we found for the specific design that we were using for the boxes. The connection system also had to be a little versatile and able to pivot from design to design as we weren't totally set on a final product.



In the same kind of way, this is how we expect our connection system to function. We want to be able to be able to pick up our box in the same way; however, with the little budgeting, we are still trying to figure out how exactly we will accomplish this task with the parts that we can find or purchase on our own.